

Received 6 July 2026, accepted 30 July 2026, date of publication 4 August 2026, date of current version 12 August 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3719781

RESEARCH ARTICLE

A Study on the Effectiveness of Gamification in STEM Higher Education: A Case Study of Probability and Mathematical Statistics

CHUAN ZHANG^{ID}, WENJIE WU^{ID}, AND ZHIMIN YANG

School of Mathematical Sciences, Qufu Normal University, Qufu, Shandong 273165, China

Corresponding author: Chuan Zhang (zhangchuan@qfnu.edu.cn)

This work was supported in part by the Undergraduate Education Reform Project of Shandong Province under Grant Z2024259, in part by the Natural Science Foundation of Shandong Province under Grant ZR2024QF012, and in part by the National Natural Science Foundation of China under Grant 62003189.

ABSTRACT To address the issues of abstract and dry instruction, insufficient interaction, and a disconnect from STEM practice in college-level Probability and Mathematical Statistics courses, this study, guided by STEM educational principles, constructs a gamified teaching system comprising motivational, mechanism, and component layers, and conducts a quasi-experimental study. A valid sample of 136 students from the School of Mathematical Sciences at Qufu Normal University was selected and divided into an experimental group (gamified instruction) and control group (traditional instruction). A four-week intervention was conducted using “Random Variables and Their Distributions” as instructional content. The results indicated that the experimental group significantly outperformed the control group across all dimensions, including learning performance, classroom participation, learning interest, learning motivation, and self-efficacy. Therefore, gamified instruction can effectively stimulate learning motivation, enhance learning performance, and improve learning attitudes, thereby meeting the reform needs of foundational mathematics and statistics courses in STEM.

INDEX TERMS Gamification, learning performance, learning motivation, probability and mathematical statistics, self-efficacy, STEM.

I. INTRODUCTION

With the acceleration of global technological progress and economic development, countries are placing greater emphasis on cultivating advanced technology and innovative talent and elevating STEM education to the level of a national strategy [1]. The mathematics curriculum system in Chinese universities typically comprises of three major categories: Advanced Mathematics, Linear Algebra, and Probability and Mathematical Statistics. These three courses serve as foundational courses for majors in the disciplines of science, engineering, management, and economics in higher education institutions, playing an irreplaceable role in scientific research, students’ further studies, and career development. Traditional university mathematics courses emphasize the transmission of theoretical knowledge, within a relatively

independent disciplinary framework. In terms of content, they prioritize the integrity of the internal logical structure of mathematics, resulting in an insufficient connection to practical applications. Regarding teaching methods, a teacher-centered model remains prevalent, lacking the cultivation of students’ innovative thinking and practical skills. The current state of curriculum struggles to meet modern society’s demand for talent possessing interdisciplinary thinking, innovative capabilities, and practical skills, making reform and restructuring urgently needed [2].

The demand for STEM professionals across all industries worldwide is continuously increasing [3]. In response to this talent shortage, various universities are constantly innovating teaching methods, aiming to stimulate students’ interest in science and technology-related majors, reduce student attrition, and comprehensively improve their academic performance [4], [5], [6], [7]. Game-based teaching, which involves embedding various game design elements in

The associate editor coordinating the review of this manuscript and approving it for publication was S. Chandrasekaran^{ID}.

non-entertainment scenarios, has become an important means of optimizing STEM subject teaching and enhancing student learning engagement and academic performance [8], [9]. Looking at the existing results, most game-based empirical studies in the STEM field are limited to computer and engineering majors, while research in other science and technology fields is still insufficient [10]. Huang et al. [11] also mentioned that various intermediary variables can influence the final effect of game-based interventions. Therefore, it is necessary to optimize the combination of game elements and conduct research with intermediary variables to create teaching plans that have lasting effects and can effectively improve learning quality. In addition, qualitative research can explore learners' thoughts and demands, effectively complementing the shortcomings of purely quantitative research. Therefore, this study adopts a mixed research design, selects the Probability Theory and Mathematical Statistics course for the experiment, and explores the effect of game-based teaching on students' grades, which sets learning participation, interest, motivation, and self-efficacy as intermediary variables, and simultaneously includes gender and previous GPA as covariates in the analysis.

As an innovative teaching method that integrates game elements and thinking into non-game contexts, gamification has garnered widespread attention and application in the field of STEM education in recent years, and has emerged as an effective solution to address the shortcomings of traditional STEM instruction, such as insufficient interactivity and lack of motivation [12]. By incorporating core elements such as points, levels, badges, instant feedback, and team competitions, dry knowledge acquisition is transformed into a challenging and engaging immersive experience. This approach effectively aligns with the core principles of STEM education, learning by doing and learning through play, stimulating students' intrinsic motivation, and enhancing classroom interaction and practical engagement.

Given the current context and existing research gaps, this study adopts a STEM education orientation and uses the Probability and Mathematical Statistics course as a vehicle to construct a three-tier gamified teaching system comprising motivational, mechanism, and component layers, and employs a quasi-experimental design [13], [14]. Using a mixed-methods approach combining quantitative testing, questionnaires, and semi-structured interviews, this study conducted a four-week instructional intervention with 136 undergraduate students at Qufu Normal University. It systematically examines the impact of gamified instruction on learning performance, classroom engagement, learning interest, learning motivation, and self-efficacy, thereby providing theoretical foundations and practical solutions for reforming foundational STEM mathematics and statistics courses in higher education [5], [15], [16].

This study's innovations are primarily reflected in three aspects: first, theoretical integration and innovation, deeply integrating gamification design, self-determination theory, self-efficacy theory, and STEM education to form a

gamified teaching theoretical framework suitable for abstract mathematical and scientific courses, thereby addressing the shortcomings of existing research on theoretical integration [4], [17]. Second, innovation in system construction: centered on the core knowledge point of random variables and their distributions, a structured system was established encompassing motivation stimulation, mechanism operation, and component implementation. This achieves a deep integration of gamification elements with knowledge points and STEM practical scenarios, addressing issues of superficial design and disjointed instruction [18], [19]. Third, innovation in research methodology: We employed a rigorous quasi-experimental design to control for extraneous variables such as teachers, content, and class duration. Through pre-test homogeneity tests, independent sample t-tests, paired t-tests, effect size analysis, and triangulation with qualitative coding, we validated teaching outcomes with multidimensional precision, yielding more scientific and reliable conclusions [20].

II. CONCEPTS AND THEORETICAL FRAMEWORK

To ensure clear research boundaries, consistent variable definitions, and scientific and operational feasibility in measurement and analysis, this section provides clear, standardized, and operational definitions for three core concepts—gamification, instructional gamification design, and teaching effectiveness in Probability and Mathematical Statistics—based on a review of authoritative domestic and international literature.

A. GAMIFICATION

Gamification is the most widely used and relatively uniformly defined core concept of educational technology and instructional design. Gamification refers to the use of game design elements in non-game contexts. In the context of higher education and STEM instruction, some scholars have further noted that gamification is essentially a motivational intervention design and behavioral guidance mechanism. The structural and systematic application of game elements stimulates learners' intrinsic motivation and sustained engagement, thereby improving the learning process and enhancing learning outcomes [17].

B. DESIGN OF GAMIFIED INSTRUCTION

The design of instructional gamification represents a design paradigm that concretizes and integrates into the curriculum, and makes the concept of gamification actionable within the educational system. Based on a synthesis of domestic research on higher education reform, this study defines it as a systematic instructional design approach that centers on course objectives, knowledge structures, and learner characteristics. It involves the systematic integration of gamification elements, operational mechanisms, and incentive rules throughout the teaching process, including classroom instruction, pre-class preparation, and post-class exercises, to construct a learning experience that is engaging, challenging, educational, and controllable, thereby enhancing

learning motivation, classroom participation, and knowledge retention [18].

C. TEACHING OUTCOMES IN PROBABILITY AND MATHEMATICAL STATISTICS

In this study, the teaching effectiveness of Probability and Mathematical Statistics refers to the observable, measurable, and comparable comprehensive changes and improvements in students' performance, engagement, and attitudes toward learning under the influence of a specific instructional intervention (gamified teaching).

III. RESEARCH DESIGN

A. RESEARCH PARTICIPANTS

This study was conducted at the Qufu Normal University, a public university in China. Informed consent and ethical approval were obtained from the participants. A total of 138 students from Classes 1 to 4 of the 2024 undergraduate teacher education cohort in the School of Mathematical Sciences were selected. We explicitly stated that participants were assigned via pre-set administrative intact classes instead of random assignments. Classes 1 and 2 served as the experimental group (73 students; boys $n=27$, girls $n=46$), whereas Classes 3 and 4 served as the control group (65 students; boys $n=21$, girls $n=44$). The average GPA of the experimental group was 3.58, while that of the control group was 3.51. Both the experimental and control groups were taught by the same instructor; the control group received instruction using traditional methods, while the experimental group's instruction incorporated gamification elements [21].

B. RESEARCH CONTENT

The textbook selected for this study is the third edition of Tutorial on Probability and Mathematical Statistics, authored by Professor Mao Shisong and published by the Higher Education Press. Owing to time constraints, the researcher focused solely on the content related to Chapter 2, "Random Variables and Their Distributions."

C. RESEARCH HYPOTHESES

Hypothesis H: Gamified design can significantly improve the teaching effectiveness of Probability and Mathematical Statistics [21].

Sub-hypotheses:

H1-1: The post-test scores of students in the experimental class were significantly higher than those in the control class.

H1-2: The improvement in scores for students in the experimental class was significantly greater than that in the control class.

H2-1: The level of classroom engagement among students in the experimental group was significantly higher than that in the control group.

H2-2: The quality of homework completion among students in the experimental group was significantly higher than that of students in the control group.

H3-1: Scores for interest in the course among students in the experimental class were significantly higher than those in the control class.

H3-2: The cores for learning motivation among the students in the experimental group were significantly higher than those in the control group.

H4: Self-efficacy scores for learning Probability and Mathematical Statistics among students in the experimental class were significantly higher than those in the control class.

D. RESEARCH INSTRUMENTS

This study employed a combination of quantitative and qualitative research tools designed to assess the teaching effectiveness and learning status of Probability and Mathematical Statistics courses. All tools were developed by referencing established domestic and international scales and course teaching standards, and were subjected to preliminary surveys to test their reliability and validity, ensuring their scientific rigor, applicability, and relevance. The specific research tools were as follows:

1) LEARNING PERFORMANCE TEST

The learning performance test consisted of a pre-test and a post-test. It serves as the core quantitative tool for measuring students' mastery of knowledge regarding random variables and their distributions, as well as for evaluating the impact of gamified instruction on learning performance. It was developed in strict accordance with the course syllabus and assessment requirements for foundational courses in science and engineering disciplines at the university level.

Design Basis and Principles: The test content closely aligns with the core knowledge points of random variables and their distributions. The pre-test focused on the foundational knowledge of probability and statistics from high school mathematics and preparatory knowledge for university courses, serving to verify the homogeneity of the initial learning foundations between students in the experimental and control groups. The post-test covered all instructional content within the experimental course cycle, random variables and their distributions, expected value of a random variable, variance and standard deviation of a random variable, common discrete distributions, common continuous distributions, and distributions of functions of random variables, comprehensively assessing students' ability to understand concepts, apply formulas, perform calculations, and analyze simple case studies. Both test papers adhered to the principles of equivalent difficulty, corresponding content, and consistent question types to avoid experimental bias caused by differences in the test papers. The difficulty ratio of the questions was set as follows: intermediate: advanced = 6:3:1, in accordance with the university course assessment standards.

Exam Structure and Scoring: Both exams were worth a total of 100 points, with a uniform duration of 90 minutes. The question types are consistent: multiple-choice questions assess the ability to distinguish basic concepts, fill-in-the-blank questions assess formula recall and simple calculations,

TABLE 1. Survey questionnaire.

Dimension	Example Item Content
Learning Engagement	1. I actively participate in classroom interactions 2. I am able to keep up with the pace of the class and actively respond to the teacher's questions 3. I proactively discuss difficult course concepts with my classmates
Interest in Learning	1. Studying the Probability and Mathematical Statistics course is an enjoyable experience for me 2. I do not find the Probability and Mathematical Statistics course boring 3. I am willing to spend time studying Probability and Mathematical Statistics on my own
Motivation	1. I study Probability and Mathematical Statistics because I am interested in the subject matter itself 2. Completing assignments in class gives me a sense of accomplishment and motivates me to continue learning 3. I hope to master probability and statistics through my studies to solve real-life problems
Self-Efficacy	1. I believe I can master the core concepts of Probability and Mathematical Statistics 2. When faced with challenging course material, I am confident that I can solve it through hard work 3. I can independently complete course-related computational and analytical problems

and open-ended questions assess the application of knowledge points and adherence to standard problem-solving procedures. The first test focused on foundational knowledge, while the second emphasized specialized course content. Both exclude obscure or unusual questions, focusing instead on core course objectives to ensure that the test results accurately reflect student learning performance.

Reliability and Validity Testing: A pre-test was administered to 25 students from non-experimental classes at the same grade level, and reliability and validity analyses were conducted using the SPSS software (version 26.0). The results showed that the Cronbach's α coefficient for the pre-test was 0.821 and that for the post-test was 0.836; both values exceeded 0.8, indicating good internal consistency reliability for the test. Content validity was evaluated by two university mathematics faculty members and an educational research expert. They concluded that the test content fully covered the knowledge points of the corresponding grade level and was highly aligned with the course learning objectives. Content validity met the required standards, and the test was deemed suitable for formal use in the experimental study.

2) TEACHING EFFECTIVENESS SURVEY QUESTIONNAIRE

The survey questionnaire is a self-administered scale and serves as the core tool for collecting quantitative data on qualitative aspects such as student learning engagement, interest, motivation, and self-efficacy [21], [22]. It was developed by referencing established scales for learning motivation and self-efficacy, as well as questionnaires on the teaching effectiveness of college mathematics courses, and was revised to suit the specific characteristics of this study. It was used to quantitatively compare the differences in student learning states between the experimental and control groups.

Questionnaire Dimensions and Item Design: The questionnaire employs a 5-point Likert scale, ranging from "inconformity" to "extreme conformity" and scored from 1 to 5, respectively. Higher scores indicate better

performance on the corresponding dimension. No reverse-coded items in the whole questionnaire, and all items adopted positive 1–5 Likert score. The composite score for each dimension was computed as the arithmetic average of all the valid item scores within that dimension. It comprises four dimensions and 28 items, as detailed in Table 1. The full questionnaire text, item-level descriptive statistics, exploratory factor loading matrix, dimension-level reliability, and convergent and discriminant validity evidence are provided in the Appendix. Discriminant validity results prove that the four constructs are empirically separable. In addition, three open-ended questions were included at the end of the questionnaire to collect students' subjective evaluations of gamified instruction versus traditional instruction as well as suggestions for improvement, thereby supplementing the quantitative data with qualitative information.

Questionnaire Reliability and Validity Testing: A total of 60 questionnaires were distributed during the preliminary survey, with 56 valid responses collected, resulting in a response rate of 93.3%. Reliability analysis showed that the overall Cronbach's α coefficient for the questionnaire was 0.872, with Cronbach's α coefficients for each dimension ranging from 0.814 to 0.853, indicating good internal consistency and stability of the questionnaire. Validity analysis employed exploratory factor analysis. The KMO value was 0.841, and Bartlett's sphericity test was <0.001 , indicating that the questionnaire has good structural validity and can be formally used for data collection in the study.

3) INTERVIEW GUIDELINES

The interview guidelines consisted of student and teacher interview guides. Qualitative research tools are used to uncover in-depth information not captured by the questionnaire or academic performance data, supplementing quantitative research findings to comprehensively understand the implementation effects, existing issues, and influencing factors of gamified teaching [21]. All interview contents were

TABLE 2. Student interview outline.

Student Interview Questions
1. Have you encountered gamified teaching models in the classroom before? What were your initial impressions upon participating in a gamified class for the first time?
2. Compared to traditional teaching, what advantages and disadvantages do you think gamified teaching has in helping students understand abstract concepts?
3. How do gamification elements such as points, levels, and leaderboards affect your motivation to learn and your classroom participation?
4. Do your mathematical background and gaming experience affect your ability to adapt to gamified instruction?
5. Have issues such as distracted attention or prioritizing games over knowledge arisen in gamified classrooms? What are the reasons for this?
6. Do you think the current gamified instructional design and activity planning are reasonable and aligned with the course requirements?
7. Has your attitude toward learning and your study methods changed as a result of gamified instruction? If so, how specifically?
8. What specific suggestions do you have for optimizing gamified instruction in Probability and Mathematical Statistics?

TABLE 3. Teacher interview guide.

Teacher Interview Questions
1. When designing gamified instruction, how do you balance gamification elements with the teaching of key concepts? How well do the two complement each other?
2. Compared to traditional teaching, what challenges does gamified instruction present in terms of pacing, discipline management, and teacher-student interaction?
3. From a practical perspective, what are the actual effects of gamified instruction on student engagement, knowledge retention, and learning efficiency?
4. Does gamified instruction increase the workload for lesson preparation and classroom organization? How does it affect the teaching schedule?
5. How do factors such as students' prior knowledge, class hours, and game formats affect the effectiveness of gamified instruction?
6. What do you consider to be the feasibility, advantages, and limitations of promoting gamified teaching in college mathematics courses?
7. Given current teaching challenges, in what ways do you think the gamified design of Probability and Mathematical Statistics courses could be optimized?
8. Do you have any other opinions or suggestions regarding the implementation of gamified teaching reforms in university mathematics courses?

approved in advance by the participants, and the entire process was handled anonymously to protect personal privacy.

Student interviews employed stratified sampling, selecting four students from each experimental class with excellent, average, and poor academic performance for semi-structured interviews. The outline centered on the design of gamified teaching experiences with specific questions is shown in Table 2. Through these interviews, we organized the subjective experiences of students at different academic levels and analyzed the actual effects and potential issues of gamified teaching on students. The teacher interviews involved an experimental course instructor and two senior mathematics educators from universities, totaling three participants. The interview guide focused on the design of the teaching implementation process, with specific questions as shown in Table 3. From the teachers' perspective, the analysis examined the implementation challenges and practical value of gamified instruction, providing qualitative support for the analysis of influencing factors and development of optimization strategies.

The interviews were conducted in one-on-one sessions, lasting 15–20 minutes per participant. The entire process was audio-recorded, and key information was recorded in real time. Transcriptions were completed within 24 hours of the interview, and the content was coded and organized to ensure the authenticity and integrity of the qualitative data. The data were subsequently cross-validated with quantitative data to enhance the persuasiveness of the research conclusions.

E. RESEARCH PROCESS

The gamified teaching intervention in this study spanned a 4-week period. The entire experiment followed a quasi-experimental design consisting of a pre-test, intervention, and post-test [21], [23]. Four strict control measures were implemented to eliminate confounding bias. First, a double blind evaluation was performed. The pre-test and post-test papers were anonymized. Two teachers scored papers independently with unified criteria and resolved disputed answers collaboratively, while the data entry staff remained blind to group allocation. Second, the teaching-related conditions

were consistent between the two groups. Identical class hours, homework assignments, and assessment arrangements were ensured. The teaching method constituted the sole manipulated variable without extra tutoring or additional guidance. Third, intervention tasks were separated from post-tests. In-class gamified activities focused on basic knowledge, whereas post-tests designed by an uninvolved third-party teacher contained exclusive comprehensive items with no repetitive exercises. Fourth, additional incentives are ruled out. In class points and badges only provide psychological achievement without counting toward final grades or offering material rewards, excluding external tangible incentives. The specific implementation steps are as follows:

1) 1 WEEK PRIOR TO INTERVENTION: PREPARATION PHASE

The gamified teaching rules, guidelines for using gamified elements, and instructional schedule were distributed to students in the experimental group via the SuperStar Learning platform. The methods for acquiring and applying scenarios of gamified elements, such as points, levels, badges, and leaderboards, are clearly explained. Simultaneously, the purpose and process of the experiment were explained to all students to eliminate cognitive biases.

Teaching tools were prepared: final versions of the pre-test and post-test papers, as well as the gamified teaching effectiveness survey, were finalized, and interview outlines were designed. An interactive gamified classroom environment was set up for the experimental group, and offline gamification components such as point-tracking sheets, achievement badges, and level-up task cards were prepared. Concurrently, dedicated modules for online point tracking, leaderboard updates, and post-class level-up task submissions were established on the SuperStar Learning platform.

Instructors completed lesson planning for gamified instruction, clearly defined the integration points between gamified elements and key learning objectives for each class session, and developed a plan to manage the classroom pace. This ensured that the gamified activities served instructional goals and avoided prioritizing gameplay over teaching.

2) WEEK 1: PRE-TEST PHASE

Students in both the experimental and control groups took a pre-test in a unified classroom setting. The exam lasted 90 minutes and was scored out of 100 points. The test content focused on foundational high school probability and statistics as well as prerequisite knowledge for the course. This was intended to verify the homogeneity of the initial knowledge base between the two groups. After the test, the papers were graded, and the scores tallied uniformly.

After the pre-test, both groups began introductory instructions on random variables and their distributions. The control group proceeded directly to the traditional lecture-based instruction, while the experimental group incorporated a gamified introductory session. The first points were awarded upon completing check-ins for foundational knowledge

points, allowing students to familiarize themselves with the gamified operational process.

3) WEEKS 2–3: GAMIFIED INTERVENTION PHASE

Both groups engaged in identical course activities, including mandatory tasks, such as homework, group projects, and in-class quizzes, as well as optional tasks, such as self-assessment tests and knowledge extension exercises. The core difference lies in the fact that all teaching activities in the experimental group incorporated a three-tier gamification design: motivation, mechanism, and component layers. Detailed explanations of each level of the framework are presented in Table 4. The specific gamification implementation design is as follows:

Motivation Layer: By creating practical STEM scenarios, students' curiosity and willingness to participate were stimulated. Key concepts were integrated with real-world application scenarios, allowing students to perceive the practical value of probability and statistical knowledge.

Mechanism Layer: Three core mechanisms were integrated: autonomous choice, a blend of competition and collaboration, and instant feedback. This approach balanced the students' sense of autonomy with their engagement.

Component Layer: The system utilizes four core components: points, achievement badges, tiered leaderboards, and challenge task cards [24], [25]. All the components were designed based on the core concept of the random variable and its distribution.

4) WEEK 4: POST-TEST AND DATA COLLECTION PHASE

The class was organized into two groups of students who completed the post-test on learning performance. The exam lasted 90 minutes and was scored out of 100 points. The test content covered all core knowledge points taught over four weeks. Papers were graded and scores tallied uniformly to calculate the improvement in scores for both groups.

After the post-test, gamified teaching effectiveness questionnaires were distributed to both groups of students on-site. Standardized instructions were provided to explain the completion requirements. The students answered the questionnaires independently, and were immediately collected. Valid questionnaires with no missing answers, no repetitive or random responses, and consistent logic were screened.

Semi-structured interviews were conducted during non-class hours; four high-performing, four average-performing, and four low-performing students were selected from the experimental group based on their scores for student interviews. Interviews were also conducted with the course instructor and two senior mathematics educators from universities. All interviews were audio-recorded, and transcription and coding were completed within 24 hours.

IV. DATA ANALYSIS

This study strictly adhered to the analytical logic of mixed-methods research, treating quantitative data analysis as the core basis for validating the effectiveness of gamified

TABLE 4. Detailed visual explanations for each level of the framework.

Levels	Core Functions	Specific Implementation Design	Alignment with Course Content
Motivation Layer	Addresses the issues of traditional math being dry and disconnected from real-world applications, stimulating intrinsic motivation to learn	Developing practical STEM case studies that integrate discrete and continuous distributions, expected values, and variance into real-world problems	Random variables, common distributions, numerical characteristics
Mechanism Level	Balancing a sense of autonomy with a sense of participation, standardizing the order of gamified classrooms, and ensuring the logical flow of classroom operations	Tiered tasks with self-selection; individual competition + group collaboration; intra-group cooperation + inter-group competition; instant scoring feedback for answers and assignments	Core knowledge points covering all chapters, with exercises that balance foundational, advanced, and extension content
Component Layer	Concrete incentive mechanisms that enable visual rewards	Points, achievement badges, leaderboards, and knowledge-based challenge task cards; a combination of the online Learning Pass platform and offline task cards	Complete coverage of random variables and their distributions

instruction, while using qualitative data analysis as a crucial supplement to interpret the causes of the effects and uncover underlying issues. These two methods mutually corroborated each other and progressed in a step-by-step manner. Quantitative analysis utilized SPSS 26.0 statistical software to perform data cleaning, descriptive statistics, and inferential statistics. Qualitative analysis employed the NVivo 11 software to code, categorize, and construct frameworks for the interview transcripts. All statistical significance tests in this study were two-tailed, and the confidence interval was 95%. All the analytical steps complied with the scientific standards of educational empirical research. The specific analytical methods and operational details are as follows:

A. QUANTITATIVE ANALYSIS

The quantitative analysis focused on academic performance test scores and teaching effectiveness survey data as its core subjects. After data preprocessing, homogeneity tests and difference tests were conducted sequentially to ensure the accuracy and validity of the results. The specific steps and procedures are as follows.

1) DATA PREPROCESSING

Outlier Handling: The 3σ rule was applied to screen for outliers in the test and questionnaire scores. If a data value fell outside the range of ± 3 standard deviations from the mean, it was assessed in conjunction with the actual response context to determine whether it was an outlier. This study identified three outliers in the questionnaire data and two outliers in the test scores, all of which were excluded. The final valid test score data set comprised 136 entries, 72 from the experimental group and 64 from the control group; the valid questionnaire data set comprised 135 entries, 71 from the experimental group, and 64 from the control group.

Reliability and Validity Validation: Reliability tests using Cronbach's α coefficient and validity tests using the KMO test and Bartlett's sphericity test were conducted on the questionnaire data to verify the scale's internal consistency and construct validity. Item analysis was performed on the score data to assess the discrimination and difficulty of the test items, ensuring that the assessment tool accurately reflected students' mastery of the subject matter.

2) INFERENCE STATISTICAL ANALYSIS

Independent-samples t-test: This was used to examine between-group differences between the experimental and control groups in pre-test scores, post-test scores, improvement in scores, and scores on each dimension of the questionnaire, thereby verifying the difference in effectiveness between gamified and traditional instruction; Prior to testing. The Levene test for homogeneity of variances is conducted. If $p > 0.05$, the assumption of homogeneity of variances was met and the results of the conventional t-test were used; if $p < 0.05$, the results of the corrected t-test were used. The significance level was set at $\alpha = 0.05$. In addition to the independent samples t-test, we adopted an Analysis of Covariance (ANCOVA), taking pre-test scores as the covariate to control for baseline differences. Considering the nested structure of students within classes, we constructed a mixed-effects model to correct for the clustering effect. We reported adjusted post-test mean values, standard errors, between-group differences, p-values, Cohen's d and 95% confidence intervals for the effect sizes. All the hypothesis tests were two-tailed.

Paired t-test: Used to examine intra-group differences in pre-test and post-test scores for both the experimental and control groups to analyze improvements in knowledge mastery among students in both groups following the instructional intervention. The significance level was set at

$\alpha = 0.05$. A p-value of <0.001 indicates an extremely significant difference, <0.01 indicates a highly significant difference, and <0.05 indicates a significant difference.

Descriptive statistical analysis: The mean (M), standard deviation (SD), maximum, minimum, and range for all quantitative data were used to visually present the distribution characteristics and degree of difference between the experimental and control groups across various indicators. The standard deviation is used to reflect the degree of dispersion of the data; a smaller standard deviation indicates a more concentrated data distribution and smaller intra-group differences.

3) EFFECT SIZE ANALYSIS

To avoid statistical bias caused by differences in sample size, the effect size d (Cohen's d) was calculated for the results of the independent sample t-test to measure the actual degree of difference between the experimental and control groups. The criteria for effect size were as follows: $d = 0.2$ indicated a small effect, $d = 0.5$ indicates a moderate effect, and $d = 0.8$ indicates a large effect. A larger effect size indicates that the actual improvement achieved by gamified teaching was more pronounced.

B. QUALITATIVE ANALYSIS

Qualitative analysis focused on transcribed student interviews, transcribed teacher interviews, and responses to open-ended questionnaire items. Using a grounded theory approach to coding, core concepts were extracted from the raw text, categorized into core categories, and an analytical framework was constructed to avoid the influence of subjective preconceptions on the results. The specific steps and procedures are as follows.

1) TEXT PREPROCESSING

Transcription and Organization: Transcribe the interview recordings verbatim into text, anonymize sensitive information such as student names and teacher details, and annotate interviewees' basic information. We categorized and organized the responses to the open-ended questionnaire items, initially grouping them into positive feedback, negative feedback, and suggestions for improvement.

Text import and coding preparation: The organized transcripts and open-ended questionnaire responses were imported into NVivo 11 software. Three node libraries—student interviews, teacher interviews, and open-ended questionnaire responses—were created to prepare for subsequent coding.

2) THREE-LEVEL CODING ANALYSIS

Open-Ended Coding: A detailed, sentence-by-sentence, and paragraph-by-paragraph interpretation of all texts to extract core concepts related to the effectiveness and implementation of gamified teaching. Repeated or similar concepts were merged and named to form initial codes; 86 initial codes were extracted in this study.

Axial Coding: Based on the intrinsic logical relationships among concepts, the initial codes were categorized and

integrated to form core categories, with the connotations and boundaries of each core category clearly defined. Through axial coding, this study consolidated the 86 initial codes into seven core categories: improvement in learning experience, changes in learning behavior, rationality of gamification design, teachers' implementation capabilities, individual student differences, adaptability of knowledge points, and issues in instructional implementation. Each core category comprised several interrelated initial codes.

Selective Coding: The core category "Formation of Gamified Teaching Effects" was extracted from the seven core categories. The causal relationships and pathways of influence between this core category and others were mapped to construct a qualitative analysis framework.

3) RELIABILITY AND VALIDITY TESTING

Reliability Testing: Two researchers independently completed open, axial and selective coding of all interview transcripts without communication, leading to coding inconsistencies. The coders then checked the conflicting items paragraphly and compiled a complete list of discrepancies. Relying on the original interview content and the study's analytical framework, they held joint discussions to unify code labels, merge redundant items and correct misclassifications, reaching full consensus on all disputed segments. A standardized unified coding dataset was formed. Cohen's Kappa coefficient was computed, with the final value reaching 0.89, indicating excellent inter-coder reliability. Finally, two university mathematics education experts reviewed the entire coding framework to reduce subjective bias and to ensure alignment with the research theme.

Validity Testing: Triangulation was employed to cross-validate the coding results from student interviews, teacher interviews, and open-ended questionnaire responses. If texts from different sources support a particular conclusion, this indicates that the conclusion is valid. In addition, two experts in mathematics education from universities were invited to review the coding framework and analytical conclusions. Based on their feedback, revisions and refinements were made to ensure scientific rigor and objectivity of the analysis results.

C. RESULTS

This study conducted a data analysis across four dimensions: pre-test homogeneity testing, learning performance, learning process, learning attitudes and psychological cognition [26], [27], [28]. The quantitative and qualitative results corroborated one another, comprehensively validating the effectiveness of gamified instruction. All statistical tests were performed with a significance level of $\alpha = 0.05$. The specific results are as follows.

1) HOMOGENEITY TEST OF PRE-TEST SCORES

To ensure the validity of the experimental results and to eliminate the interference of students' initial knowledge differences on the teaching effect, Levene's test for variance homogeneity, independent sample t-test and Analysis of

TABLE 5. Results of homogeneity tests ($M \pm SD$, out of 100).

Group	Valid Sample Size	Mean (M)	Standard Deviation (SD)	Maximum	Minimum	Levene's test (F/p)	t-value (df)	p-value	95% Confidence Interval
Experimental group	72	62.58	7.82	78.00	45.00	0.523/0.471	0.352 (134)	0.725	[-1.89, 3.11]
Control group	64	61.97	8.15	76.00	43.00	-	-	-	-

TABLE 6. Inter-group balance test of all baseline covariates.

Baseline Indicator	Index	Experimental Group	Control Group	Test Statistic	p-value	Balance Judgment ($p \geq 0.05$ balanced)
Gender	Male, n(%)	27(37.50%)	21(32.81%)	$\chi^2=0.326$	0.568	Balanced
	Female, n(%)	46(62.50%)	44(67.19%)			
Prior GPA	$M \pm SD$	3.58 ± 0.43	3.51 ± 0.47	$t=0.927(134)$	0.355	Balanced
Pre-test Score	$M \pm SD$	62.58 ± 7.82	61.97 ± 8.15	$t=0.352(134)$	0.725	Balanced

Table Note: All baseline covariates show no significant inter-group imbalance.

TABLE 7. ANCOVA results for post-test achievement (Controlling pre-test scores).

Source	Type III Sum of Squares	df	Mean Square	F	p	Partial η^2
Pre-test (Covariate)	2647.35	1	2647.35	76.42	<0.001	0.365
Group (Experimental vs Control)	3215.68	1	3215.68	92.75	<0.001	0.412
Error	4589.21	132	34.77	—	—	—

Covariance were conducted on the pre-test scores of learning performance of the experimental group and the control group. The results are presented in Table 5, Table 6 and Table 7.

As shown in Table 5, the results of the Levene test for homogeneity of variances indicate $F = 0.523$ and $p = 0.471 > 0.05$, suggesting that the pre-test scores of the experimental group and the control group satisfy the assumption of homogeneity of variances, and thus a conventional independent-samples t-test can be used. The results of the independent samples t-test showed that $t = 0.352$, $df = 134$, and $p = 0.725 > 0.05$. Furthermore, the 95% confidence interval included 0, indicating that there was no statistically significant difference in foundational knowledge of Probability and Mathematical Statistics between the experimental and control groups at the initial stage of the experiment. However, the analysis of covariates showed that the covariate $F = 76.42$, with $p < 0.001$, indicating that the pre-test scores have an impact on the post-test scores. It is necessary to use ANCOVA to control the baseline. After controlling the students' original foundation, the independent variable was $F = 92.75$, with $p < 0.001$. This indicates that even after excluding the differences in students' foundation, gamification teaching can improve test scores. The

means, standard deviations, maximum values, and minimum values of the pre-test scores for both groups were similar, and the degrees of dispersion within each group were comparable. This further indicates that the students' initial learning foundations were homogeneous, the experimental variable was effectively established, and any differences in the effects of subsequent instructional interventions could be attributed to the application of the gamified teaching model.

2) ANALYSIS OF THE LEARNING PERFORMANCE DIMENSION

This study analyzed the effectiveness of gamified instruction in enhancing students' knowledge mastery from three perspectives: post-test scores, improvement in scores, and pre-post differences within groups [29], [30]. Paired-sample t-tests, independent-sample t-tests, and effect size tests were employed, and the results are presented in Tables 8 and 9.

As shown in Tables 8 and 9, the results of the paired-sample t-tests for both the experimental and control groups were $p = 0.000 < 0.001$, indicating that both teaching models improved students' knowledge mastery and achieved basic teaching objectives. However, the mean difference for the experimental group was significantly greater than that of

TABLE 8. Comparison of post-test scores and score improvement (M $\pm$ SD, out of 100).

Indicator	Experimental Group (n=72)	Control Group (n=64)	Levene's Test (F/p)	t-value (df)	p-value	Effect size d	95% confidence interval
Post-test score	78.64 $\pm$ 6.93	66.28 $\pm$ 7.56	1.025/0.313	9.283(134)	<0.001	1.61	[9.87, 15.05]
Improvement in scores	16.06 $\pm$ 5.21	4.31 $\pm$ 4.89	0.856/0.356	12.564(134)	<0.001	2.20	[9.98, 13.52]

TABLE 9. Results of the paired-sample t-test for pre- and post-test scores.

Group	Difference in means (post-test vs. pre-test)	Standard Deviation (SD)	t-value (df)	p-value	Significance of difference
Experimental group	16.06	5.21	26.852 (71)	<0.001	Extremely significant
Control group	4.31	4.89	7.126(63)	<0.001	Extremely significant

TABLE 10. Comparison of learning process dimensions between the experimental and control groups.

Indicator Type	Specific Indicator	Experimental Group (M $\pm$ SD)	Control Group (M $\pm$ SD)	t-value (df)	p-value	Effect size d
Subjective questionnaire scores	Class Participation (7 items, maximum score of 5)	4.25 $\pm$ 0.38	3.12 $\pm$ 0.45	15.628 (133)	0.000	2.75
	Homework Completion Rate (4 items, maximum score of 5)	4.31 $\pm$ 0.35	3.08 $\pm$ 0.42	16.892(133)	0.000	2.96
	Teamwork Performance (3 items, maximum score of 5)	4.18 $\pm$ 0.40	3.05 $\pm$ 0.48	14.564(133)	0.000	2.55
Classroom Objective Data	Number of times students actively answered questions in class (times/class)	8.62 $\pm$ 2.15	2.35 $\pm$ 1.08	19.853 (134)	0.000	3.48
	Homework completion rate (%)	98.63	87.21	-	<0.05	-
	Post-class optional task completion rate (%)	72.30	28.50	-	<0.05	-

the control group, suggesting that the improvement achieved through gamified instruction was significantly greater than that achieved through traditional instruction.

Regarding the post-test scores, the independent samples t-test results showed $p = 0.000 < 0.001$, with an effect size of $d = 1.61$, indicating that the experimental group's post-test scores were significantly higher than those of the control group, and the actual difference was substantial. Regarding the magnitude of score improvement, the results of the independent samples t-test showed $p = 0.000 < 0.001$, with an effect size of $d = 2.20$, indicating that gamified instruction is far more effective than traditional teaching models in improving students' learning performance.

Regarding data distribution characteristics, the median post-test score of the experimental group was significantly higher than that of the control group, indicating that the experimental group's mastery of knowledge was more concentrated, with less polarization compared to the control group. The control group's post-test scores exhibited greater

dispersion, suggesting that variations in learning outcomes were more pronounced under traditional instruction.

The above results fully validated sub-hypotheses H1-1 and H1-2, indicating that gamified instruction effectively helps students understand the core concepts of Probability and Mathematical Statistics, significantly improving their mastery of knowledge and test-taking abilities.

3) ANALYSIS OF THE LEARNING PROCESS DIMENSION

This study analyzed the differences between the experimental and control groups using an independent samples t-test across two dimensions: subjective questionnaire scores and objective classroom observation data. The analysis covered three core indicators, class participation, completion of post-class assignments, and group collaboration performance, with the results shown in Table 10. A qualitative supplementary analysis was also conducted based on classroom observation records.

TABLE 11. Comparison of scores on learning attitude and psychological cognitive dimensions ($M \pm SD$, maximum score of 5).

Dimension	Number of items	Experimental group (n=71)	Control group (n=64)	Levene's test (F/p)	t-value (df)	p-value	Effect size d
Interest in learning	6	4.32 $\pm$ 0.41	3.05 $\pm$ 0.52	1.256/0.263	14.892(133)	0.000	2.61
Motivation to Learn	8	4.18 $\pm$ 0.39	3.21 $\pm$ 0.48	0.985/0.323	12.564 (133)	0.000	2.20
Academic Self-Efficacy	7	4.26 $\pm$ 0.37	3.18 $\pm$ 0.46	0.852/0.357	13.257 (133)	0.000	2.34

As shown in Table 10 and classroom observation records, the experimental group significantly outperformed the control group on all learning process indicators ($p < 0.05$), with effect size d exceeding 2.5. This indicates that gamified instruction has a significant positive impact on students' engagement in learning behaviors, significantly enhancing their classroom participation, post-class assignment completion rates, and group collaboration skills. The difference in the completion rate of optional post-class tasks was particularly significant, indicating that gamified design can effectively stimulate students' motivation for self-directed learning, encouraging them to actively engage in expanding their knowledge rather than merely fulfilling mandatory tasks.

These results validate sub-hypotheses H2-1 and H2-2, indicating that gamified instruction can effectively improve students' learning behaviors and enhance their engagement and effectiveness in the learning process.

4) ANALYSIS OF LEARNING ATTITUDE AND PSYCHOLOGICAL COGNITION DIMENSIONS

This study analyzed the differences in questionnaire scores between the experimental and control groups using an independent samples t-test across three core dimensions: learning interest, learning motivation, and self-efficacy for learning [31]. All dimensions employed a 5-point Likert scale (1 = inconformity, 5 = extreme conformity), with higher scores indicating better performance in the corresponding dimension [32], [33]. The results are shown in Table 11, and qualitative interpretations were conducted by combining the open-ended questionnaire items with student interviews.

As shown in Table 11 and the qualitative analysis, the experimental group scored significantly higher than the control group on all three dimensions—learning interest, learning motivation, and self-efficacy for learning ($p = 0.000 < 0.001$), with effect sizes d all exceeding 2.0. This indicates that gamified instruction can significantly improve students' learning attitudes and enhance their intrinsic learning motivation and learning self-efficacy [34], [35].

These results validate sub-hypotheses H3-1, H3-2, and H4, indicating that gamified instruction can stimulate students' learning motivation and improve their psychological state toward learning.

V. RESEARCH CONCLUSION AND OUTLOOK

A. RESEARCH CONCLUSION

This study was conducted with 136 undergraduate students at Qufu Normal University using a quasi-experimental design, and investigated the effectiveness of gamified instruction in Probability and Mathematical Statistics courses under the STEM framework. The results showed that the experimental group's pre-test scores were not significantly different from those of the control group, indicating homogeneity. After a 4-week intervention, the experimental group significantly outperformed the control group in post-test scores and the magnitude of score improvement, demonstrating a significant enhancement in learning performance. In terms of the learning process, the experimental group demonstrated significantly higher classroom participation, post-class assignment completion rates, and group collaboration performance than the control group, and exhibited a stronger willingness to engage in self-directed learning [36]. In terms of learning attitudes and psychological cognition, the experimental group scored significantly higher on the measures of learning interest, motivation, and self-efficacy. The study confirms that a gamified teaching system comprising motivational, mechanistic, and component layers can effectively stimulate learning motivation, enhance learning performance, and strengthen classroom interaction, thereby meeting the reform needs of foundational mathematics and statistics courses in STEM higher education [37], [38].

B. LIMITATIONS OF THE STUDY

Although this study validated the effectiveness of gamified instruction in Probability and Mathematical Statistics courses, certain limitations remain due to constraints such as study duration, research participants, and instructional content. These limitations are reflected in the following aspects:

First, the quasi-experimental design using whole classes rather than individual randomization introduces class clustering bias, as students within the same class are not independent; failing to fully model this with mixed-effects approaches weakens causal inference.

Second, the baseline homogeneity test was incomplete, relying mainly on pre-test scores and later adding comparisons only for gaming experience and prior exposure, without incorporating these individual differences as control variables from the start, compromising experimental rigor.

Third, the immediate post-intervention questionnaire risks novelty effects and social desirability bias, inflating short-term motivation and interest scores, while the cross-sectional design cannot capture long-term changes in intrinsic curiosity versus external stimulation.

Fourth, limitations in scope and sample homogeneity restrict generalizability: the four-week intervention covers only one chapter, lacks long-term tracking, and draws exclusively on teacher-education students from a single institution and discipline, making it difficult to assess the gamified approach's adaptability across different knowledge types, courses, institutions, and student populations [17], [20], [39], [40].

C. FUTURE RESEARCH DIRECTIONS

Building on the conclusions and limitations of this study, future research could focus on optimizing the teaching system, expanding the scope of the research, deepening the analysis, and facilitating practical implementation. This would promote the deeper and broader application of gamified design in foundational mathematics courses in STEM higher education. The specific future directions are as follows.

The duration of instructional interventions should be extended and the coverage of instructional content broadened to validate the long-term effectiveness and comprehensive adaptability of gamified teaching. In the future, gamified teaching interventions could be extended to cover the entire course of Probability and Mathematical Statistics, encompassing all the core chapters. This would allow for an analysis of the stability of gamified teaching's effectiveness over the long term, a comparison of its adaptability to knowledge points of varying difficulty and types, and the development of a more comprehensive and systematic gamified teaching system [11], [41], [42].

The scope of the research subjects should be expanded to conduct comparative studies across institutions, disciplines, and academic levels, thereby enhancing the generalizability of the research conclusions. In the future, students from universities of varying levels and types could be selected as research subjects, including students majoring in mathematics, science and engineering, business and management, and humanities. By conducting quasi-experimental studies across multiple groups, we can analyze the differences in adaptability of the gamified teaching system across different disciplines and institutions [43]. Combining these findings with the practical STEM needs of various disciplines, we can optimize gamified teaching schemes and construct a personalized gamified teaching system tailored to specific disciplines and educational levels [44].

In summary, gamified design, as an innovative teaching method, holds broad application prospects in foundational mathematics courses in STEM higher education. Future research should build upon this study to further optimize the teaching system, expand the scope of research, and deepen analytical rigor. This will promote the standardized, intelligent, and large-scale development of gamified teaching, effectively enhance the quality of foundational

courses in STEM higher education, and cultivate more innovative STEM talent equipped with interdisciplinary thinking, innovative capabilities, and practical skills. These efforts will provide strong support for China's goal of becoming an educational powerhouse and achieving self-reliance and strength in science and technology [13], [14], [18].

REFERENCES

- [1] T. K. F. Chiu, Y. Li, M. Ding, J. Hallström, and M. D. Koretsky, "A decade of research contributions and emerging trends in the international journal of STEM education," *Int. J. STEM Educ.*, vol. 12, no. 1, Feb. 2025, doi: [10.1186/s40594-025-00533-7](https://doi.org/10.1186/s40594-025-00533-7).
- [2] Y. Li, "Trends of high impact educational research in scientific disciplines," *Int. J. STEM Educ.*, vol. 13, no. 1, Apr. 2026, doi: [10.1186/s40594-026-00607-0](https://doi.org/10.1186/s40594-026-00607-0).
- [3] R. Batchelor, H. Ali, K. Gardner-Vandy, A. Gold, J. MacKinnon, and P. Asher, "Reimagining STEM workforce development as a braided river," *Eos*, vol. 102, Apr. 2021, doi: [10.1029/2021eo157277](https://doi.org/10.1029/2021eo157277).
- [4] Y. Gui, Z. Cai, Y. Yang, L. Kong, X. Fan, and R. H. Tai, "Effectiveness of digital educational game and game design in STEM learning: A meta-analytic review," *Int. J. STEM Educ.*, vol. 10, no. 1, May 2023, doi: [10.1186/s40594-023-00424-9](https://doi.org/10.1186/s40594-023-00424-9).
- [5] J. Qin, "On the reform of education methods that adapt to STEM development demand," *Int. J. Educ. Humanities*, vol. 6, no. 2, pp. 141–143, Dec. 2022, doi: [10.54097/ijeh.v6i2.3660](https://doi.org/10.54097/ijeh.v6i2.3660).
- [6] M. Videnovik, T. Vold, L. Kjønik, A. Madevska Bogd, and V. Trajkovik, "Game-based learning in computer science education: A scoping literature review," *IJ STEM Ed*, vol. 10, p. 54, Aug. 2023, doi: [10.1186/s40594-023-00447-2](https://doi.org/10.1186/s40594-023-00447-2).
- [7] M. Zeidan, X. Huang, L. Xiao, and R. Zhao, "Improving student engagement using a video-enabled activity-based learning: An exploratory study to STEM preparatory education in UAE," *J. Learn. Dev. High. Educ.*, vol. 24, pp. 1–12, 2022, doi: [10.47408/jldhe.vi24.888](https://doi.org/10.47408/jldhe.vi24.888).
- [8] S. Deterding, "Situating motivational affordances of game elements: A conceptual model," in *Proc. Gamification, Using Game Design Elements Non-Gaming Contexts, CHI Workshop*, vol. 10, 2011.
- [9] I. Silva, A. Wong, B. Auria, D. Zambrano, and V. Echeverria, "Gamification in engineering education: Exploring students' performance, motivation, and engagement," in *Proc. IEEE 6th Ecuador Tech. Chapters Meeting (ETCM)*, Ecuador, Oct. 2022, pp. 1–6, doi: [10.1109/ETCM56276.2022.9935729](https://doi.org/10.1109/ETCM56276.2022.9935729).
- [10] B. Castillo-Parra, B. G. Hidalgo-Cajo, M. Váscquez-Barrera, and J. Oleas-López, "Gamification in higher education: A review of the literature," *World J. Educ. Technol., Current Issues*, vol. 14, no. 3, pp. 797–816, May 2022, doi: [10.18844/wjet.v14i3.7341](https://doi.org/10.18844/wjet.v14i3.7341).
- [11] R. Huang, A. D. Ritzhaupt, M. Sommer, J. Zhu, A. Stephen, N. Valle, J. Hampton, and J. Li, "The impact of gamification in educational settings on student learning outcomes: A meta-analysis," *Educ. Technol. Res. Develop.*, vol. 68, no. 4, pp. 1875–1901, Aug. 2020, doi: [10.1007/s11423-020-09807-z](https://doi.org/10.1007/s11423-020-09807-z).
- [12] S. E. Ojonuba, G. Türkmen, and S. Toker, "Enhancing web development education with game-based and gamification learning: A study of engagement, motivation, and performance," *IEEE Access*, vol. 13, pp. 137048–137066, 2025, doi: [10.1109/ACCESS.2025.3595510](https://doi.org/10.1109/ACCESS.2025.3595510).
- [13] F. Kayan-Fadlilmula, A. Sellami, N. Abdelkader, and S. Umer, "A systematic review of STEM education research in the GCC countries: Trends, gaps and barriers," *Int. J. STEM Educ.*, vol. 9, no. 1, Jan. 2022, doi: [10.1186/s40594-021-00319-7](https://doi.org/10.1186/s40594-021-00319-7).
- [14] Y. Li, K. Wang, Y. Xiao, and J. E. Froyd, "Research and trends in STEM education: A systematic review of journal publications," *Int. J. STEM Educ.*, vol. 7, no. 1, Mar. 2020, doi: [10.1186/s40594-020-00207-6](https://doi.org/10.1186/s40594-020-00207-6).
- [15] H. Jiang, D. Zhu, R. Chugh, D. Turnbull, and W. Jin, "Virtual reality and augmented reality-supported K-12 STEM learning: Trends, advantages and challenges," *Educ. Inf. Technol.*, vol. 30, no. 9, pp. 12827–12863, Jun. 2025, doi: [10.1007/s10639-024-13210-z](https://doi.org/10.1007/s10639-024-13210-z).
- [16] D. Zhao, J. Playfoot, C. De Nicola, G. Guarino, M. Bratu, F. Di Salvatore, and G.-M. Muntean, "An innovative multi-layer gamification framework for improved STEM learning experience," *IEEE Access*, vol. 10, pp. 3879–3889, 2022, doi: [10.1109/ACCESS.2021.3139729](https://doi.org/10.1109/ACCESS.2021.3139729).
- [17] M. Li, S. Ma, and Y. Shi, "Examining the effectiveness of gamification as a tool promoting teaching and learning in educational settings: A meta-analysis," *Frontiers Psychol.*, vol. 14, Oct. 2023, doi: [10.3389/fpsyg.2023.1253549](https://doi.org/10.3389/fpsyg.2023.1253549).

- [18] D. T. K. Ng, C. Xinyu, J. K. L. Leung, and S. K. W. Chu, "Fostering students' AI literacy development through educational games: AI knowledge, affective and cognitive engagement," *J. Comput. Assist. Learn.*, vol. 40, no. 5, pp. 2049–2064, Oct. 2024, doi: [10.1111/jcal.13009](https://doi.org/10.1111/jcal.13009).
- [19] A. Khaldi, R. Bouzidi, and F. Nader, "Gamification of e-learning in higher education: A systematic literature review," *Smart Learn. Environ.*, vol. 10, no. 1, Jan. 2023, doi: [10.1186/s40561-023-00227-z](https://doi.org/10.1186/s40561-023-00227-z).
- [20] L. L. Lee Yan and M. E. E. M. Matore, "Gamification trend in students' mathematics learning through systematic literature review," *Int. J. Academic Res. Progressive Educ. Develop.*, vol. 12, no. 1, pp. 433–461, Jan. 2023, doi: [10.6007/ijarped/v12-i1/15732](https://doi.org/10.6007/ijarped/v12-i1/15732).
- [21] M. Ortiz-Rojas, K. Chiliza, M. Valcke, and C. Bolanos-Mendoza, "How gamification boosts learning in STEM higher education: A mixed methods study," *Int. J. STEM Educ.*, vol. 12, no. 1, p. 1, Jan. 2025, doi: [10.1186/s40594-024-00521-3](https://doi.org/10.1186/s40594-024-00521-3).
- [22] J. Chen and M. Liang, "Play hard, study hard? The influence of gamification on students' study engagement," *Frontiers Psychol.*, vol. 13, Oct. 2022, doi: [10.3389/fpsyg.2022.994700](https://doi.org/10.3389/fpsyg.2022.994700).
- [23] S. Ros, S. González, A. Robles, L. Tobarra, A. Caminero, and J. Cano, "Analyzing Students' self-perception of success and learning effectiveness using gamification in an online cybersecurity course," *IEEE Access*, vol. 8, pp. 97718–97728, 2020, doi: [10.1109/ACCESS.2020.2996361](https://doi.org/10.1109/ACCESS.2020.2996361).
- [24] E. J. Theobald et al., "Active learning narrows achievement gaps for under-represented students in undergraduate science, technology, engineering, and math," in *Proc. Natl. Acad. Sci., USA*, vol. 117, no. 12, Mar. 2020, pp. 6476–6483, doi: [10.1073/pnas.1916903117](https://doi.org/10.1073/pnas.1916903117).
- [25] M. D. Hanus and J. Fox, "Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance," *Comput. Educ.*, vol. 80, pp. 152–161, Jan. 2015, doi: [10.1016/j.compedu.2014.08.019](https://doi.org/10.1016/j.compedu.2014.08.019).
- [26] M. Sailer and L. Homner, "The gamification of learning: A meta-analysis," *Educ. Psychol. Rev.*, vol. 32, no. 1, pp. 77–112, Mar. 2020, doi: [10.1007/s10648-019-09498-w](https://doi.org/10.1007/s10648-019-09498-w).
- [27] S. Bai, K. F. Hew, and B. Huang, "Does gamification improve student learning outcome? Evidence from a meta-analysis and synthesis of qualitative data in educational contexts," *Educ. Res. Rev.*, vol. 30, Jun. 2020, Art. no. 100322, doi: [10.1016/j.edurev.2020.100322](https://doi.org/10.1016/j.edurev.2020.100322).
- [28] H. M. M. Ahmed, H. A. El-Sabagh, and D. M. Elbourhamy, "Effect of gamified, mobile, cloud-based learning management system (GMCLMS) on student engagement and achievement," *Int. J. Educ. Technol. Higher Educ.*, vol. 22, no. 1, p. 49, Aug. 2025, doi: [10.1186/s41239-025-00541-1](https://doi.org/10.1186/s41239-025-00541-1).
- [29] S. Halawa, T.-C. Lin, and Y.-S. Hsu, "Exploring instructional design in K-12 STEM education: A systematic literature review," *Int. J. STEM Educ.*, vol. 11, no. 1, Sep. 2024, doi: [10.1186/s40594-024-00503-5](https://doi.org/10.1186/s40594-024-00503-5).
- [30] I. Y. Zairon, T. S. M. T. Wook, S. M. Salleh, and H. A. Dahlan, "User model for virtual learning based on adaptive gamification," *IEEE Access*, vol. 13, pp. 24028–24040, 2025, doi: [10.1109/ACCESS.2025.3537599](https://doi.org/10.1109/ACCESS.2025.3537599).
- [31] L. Li, K. F. Hew, and J. Du, "Gamification enhances student intrinsic motivation, perceptions of autonomy and relatedness, but minimal impact on competency: A meta-analysis and systematic review," *Educ. Technol. Res. Develop.*, vol. 72, no. 2, pp. 765–796, Apr. 2024, doi: [10.1007/s11423-023-10337-7](https://doi.org/10.1007/s11423-023-10337-7).
- [32] M.-B. Ibanez, A. Di-Serio, and C. Delgado-Kloos, "Gamification for engaging computer science students in learning activities: A case study," *IEEE Trans. Learn. Technol.*, vol. 7, no. 3, pp. 291–301, Jul. 2014, doi: [10.1109/TLT.2014.2329293](https://doi.org/10.1109/TLT.2014.2329293).
- [33] L. Zahedi, J. Batten, M. Ross, G. Potvin, S. Damas, P. Clarke, and D. Davis, "Gamification in education: A mixed-methods study of gender on computer science students' academic performance and identity development," *J. Comput. Higher Educ.*, vol. 33, no. 2, pp. 441–474, Aug. 2021, doi: [10.1007/s12528-021-09271-5](https://doi.org/10.1007/s12528-021-09271-5).
- [34] M. Ortiz-Rojas, K. Chiliza, and M. Valcke, "Gamification through leaderboards: An empirical study in engineering education," *Comput. Appl. Eng. Educ.*, vol. 27, no. 4, pp. 777–788, Jul. 2019, doi: [10.1002/cae.12116](https://doi.org/10.1002/cae.12116).
- [35] C. Latorre-Coscolluela, V. Sierra-Sánchez, and S. Vazquez-Toledo, "Gamification, collaborative learning and transversal competences: Analysis of academic performance and students' perceptions," *Smart Learn. Environments*, vol. 12, pp. 1–17, Jan. 25, doi: [10.1186/s40561-024-00361-2](https://doi.org/10.1186/s40561-024-00361-2).
- [36] M. Sailer, J. U. Hense, S. K. Mayr, and H. Mandl, "How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction," *Comput. Hum. Behav.*, vol. 69, pp. 371–380, Apr. 2017, doi: [10.1016/j.chb.2016.12.033](https://doi.org/10.1016/j.chb.2016.12.033).
- [37] C. Meşe and Ö. Dursun, "Influence of gamification elements on emotion, interest and online participation," *Educ. Sci.*, vol. 43, no. 196, pp. 67–95, Oct. 2018, doi: [10.15390/eb.2018.7726](https://doi.org/10.15390/eb.2018.7726).
- [38] C. Meng, M. Zhao, Z. Pan, Q. Pan, and C. J. Bonk, "Investigating the impact of gamification components on online learners' engagement," *Smart Learn. Environ.*, vol. 11, no. 1, Oct. 2024, doi: [10.1186/s40561-024-00336-3](https://doi.org/10.1186/s40561-024-00336-3).
- [39] J. Krath, L. Schürmann, and H. F. O. von Korflesch, "Revealing the theoretical basis of gamification: A systematic review and analysis of theory in research on gamification, serious games and game-based learning," *Comput. Hum. Behav.*, vol. 125, Dec. 2021, Art. no. 106963, doi: [10.1016/j.chb.2021.106963](https://doi.org/10.1016/j.chb.2021.106963).
- [40] J. Bourgonjon, M. Valcke, R. Soetaert, and T. Schellens, "Students' perceptions about the use of video games in the classroom," *Comput. Educ.*, vol. 54, no. 4, pp. 1145–1156, May 2010, doi: [10.1016/j.compedu.2009.10.022](https://doi.org/10.1016/j.compedu.2009.10.022).
- [41] E. D. Mekler, F. Brühlmann, A. N. Tuch, and K. Opwis, "Towards understanding the effects of individual gamification elements on intrinsic motivation and performance," *Comput. Hum. Behav.*, vol. 71, pp. 525–534, Jun. 2017, doi: [10.1016/j.chb.2015.08.048](https://doi.org/10.1016/j.chb.2015.08.048).
- [42] M. Bond, K. Buntins, S. Bedenlier, O. Zawacki-Richter, and M. Kerres, "Mapping research in student engagement and educational technology in higher education: A systematic evidence map," *Int. J. Educ. Technol. Higher Educ.*, vol. 17, no. 1, Jan. 2020, doi: [10.1186/s41239-019-0176-8](https://doi.org/10.1186/s41239-019-0176-8).
- [43] P. M. Ross, E. Scanes, P. Poronnik, H. Coates, and W. Locke, "Understanding STEM academics' responses and resilience to educational reform of academic roles in higher education," *Int. J. STEM Educ.*, vol. 9, no. 1, Jan. 2022, doi: [10.1186/s40594-022-00327-1](https://doi.org/10.1186/s40594-022-00327-1).
- [44] A. J. Kleinschmit, A. Rosenwald, E. F. Ryder, S. Donovan, B. Murdoch, N. F. Grandgenett, M. Pauley, E. Triplett, W. Tappich, and W. Morgan, "Accelerating STEM education reform: Linked communities of practice promote creation of open educational resources and sustainable professional development," *Int. J. STEM Educ.*, vol. 10, no. 1, Feb. 2023, doi: [10.1186/s40594-023-00405-y](https://doi.org/10.1186/s40594-023-00405-y).



CHUAN ZHANG received the B.S. degree in statistics from Qufu Normal University, China, in 2013, the M.S. degree in applied mathematics from Nanjing University of Finance and Economics, China, in 2015, and the Ph.D. degree in computer applications technology from Dalian University of Technology, China, in 2018. He is currently an Associate Professor with the School of Mathematical Sciences, Qufu Normal University. His research interests include STEM education, artificial intelligence and mathematics education, the synchronization of complex networks, neural networks, and distributed parameter systems.



WENJIE WU was born in Qingdao, Shandong, China, in 2002. She received the B.S. degree in science from the University of Jinan, China, in 2024. She is currently pursuing the master's degree in subject teaching (mathematics) with the School of Mathematical Sciences, Qufu Normal University, China.



ZHIMIN YANG was born in Tai'an, Shandong, China, in 2003. He received the B.S. degree in science from Shandong Normal University, China, in 2025. He is currently pursuing the master's degree in subject teaching (mathematics) with the School of Mathematical Sciences, Qufu Normal University, China.

...